

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Main bearing cap puller, Part Number ST-1178
- Air impact wrench and socket, Part Numbers 3163059 and 5573072

Additional Service Items

- No additional service items required

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Cummins Inc. recommends replacing the thrust bearings when the main bearings are replaced. Use the following procedure for thrust bearing replacement. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-007.html>)

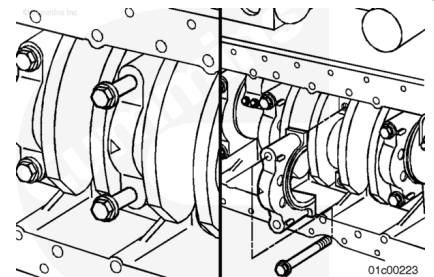
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-037.html>)

- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-035.html)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-031.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-089.html)

Remove

⚠ CAUTION ⚠

Mark any unmarked bearing caps before removing them from the cylinder block. Do not remove main bearing capscrews until the main bearing cap is pulled, or the cap could fall out, causing personal injury and parts damage.



If the crankshaft is **not** going to be removed, replace the main bearings one at a time.

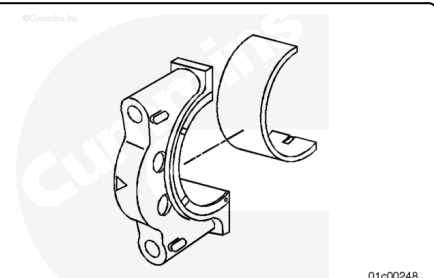
Loosen the main bearing cap capscrews.

Use the main bearing cap puller, Part Number ST-1178, to remove the cap. The tool **must** be centered on the cap.

Remove the main bearing capscrews and cap.

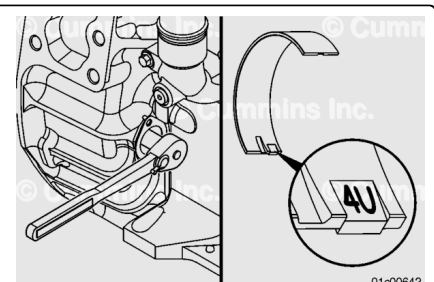
Mark the bearing shells with the journal number from which they were removed.

Remove the lower main bearing shell from the cap.



Rotate the crankshaft with the air compressor drive or a barring device to remove the main bearing shell. Refer to Procedure 000-017 in Section 0.
(/qs3/pubsys2/xml/en/procedures/377/377-000-017.html)

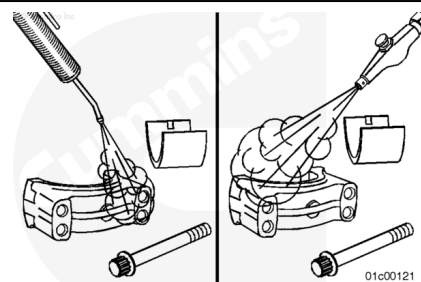
Mark each main bearing for location if they are to be reused in the engine.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



Use a soft bristle brush to clean the parts.

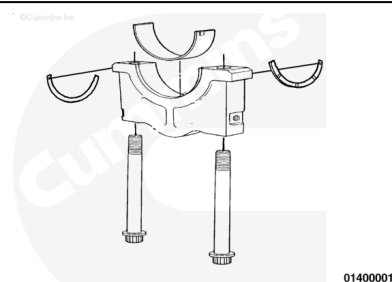
Dry the parts with compressed air.

Inspect the main bearing cap and capscrews for damage.

Inspect the bearing shells and thrust bearings for nicks, scratches, and other damage.

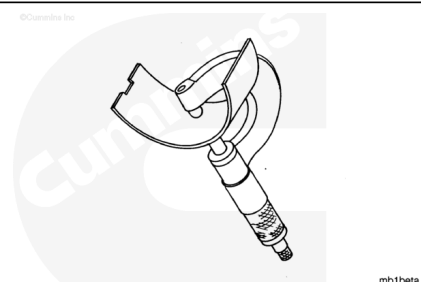
If the main bearings are damaged, inspect the crankshaft main bearing journals. If the crankshaft is damaged, the engine **must** be removed for repair. Refer to Procedure 000-001 in Section 0. (</qs3/pubsys2/xml/en/procedures/377/377-000-001.html>)

Cummins Inc. recommends replacing the thrust bearings when the main bearings are replaced. Use the following procedure for thrust bearing replacement. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-007.html>)



Use an outside diameter ball-tipped micrometer to measure the main bearing shell thickness.

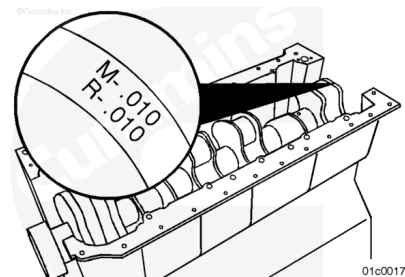
Standard Main Bearing Shell Thickness		
mm		in
3.196	MIN	0.126
3.214	MAX	0.127



If the bearing shell is **not** within specifications, it **must** be replaced.

For more detailed information on bearing damage, refer to Analysis and Prevention of Bearing Failures, Bulletin 3810387 (/qs3/pubsys2/xml/en/bulletin/3810387.html).

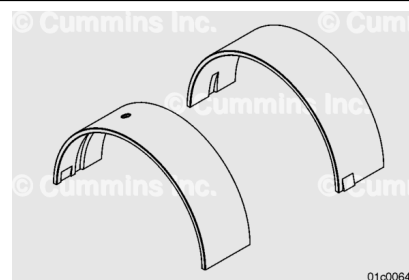
Crankshafts that are ground undersize in the connecting rod and main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.



Install

⚠ CAUTION ⚠

The cylinder block saddle and the cap mating surfaces must be clean and dry when the bearing shells are installed. Used bearings must be installed in their original location or engine damage can occur.



⚠ CAUTION ⚠

The bearing tang must fit into the slot in the bearing saddle or engine damage can occur.

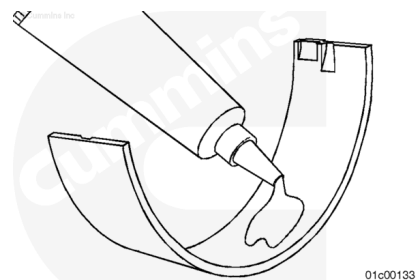
⚠ CAUTION ⚠

Do not use a capscrew or any hard metal object to install bearings, as this can damage the crankshaft and cause an engine failure.

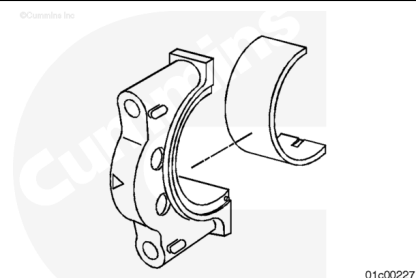
Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the upper main bearing shell to crankshaft journal mating surface.

Start the bearing into the upper saddle by hand. Slide the bearing into position. Take care to **not** damage the crankshaft journal while installing the bearing.

Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the inside diameter of the bearing shells and the liner side of the upper thrust bearings.

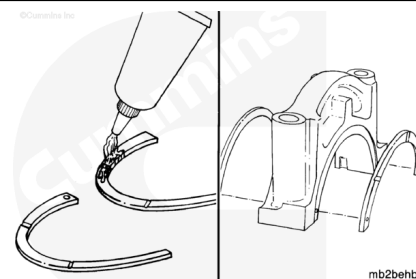


Align the lower main bearing shell and push it into place.



⚠ CAUTION ⚠

**Make sure the bevel side is toward the crankshaft journal.
Incorrect thrust bearing installation will result in engine damage.**



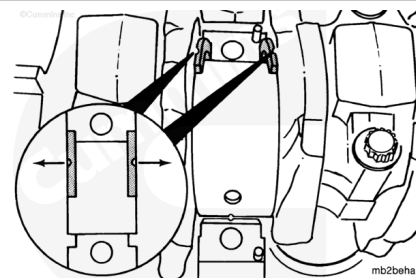
Only number 4 main bearing cap has thrust bearings.

Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the liner side of the lower thrust bearings.

Install the lower thrust bearings in the number 4 main bearing cap as illustrated.

⚠ CAUTION ⚠

**Make sure the bevel side is toward the crankshaft journal.
Incorrect thrust bearing installation will result in engine damage.**



Install the upper thrust bearings in the number 4 main bearing saddle.

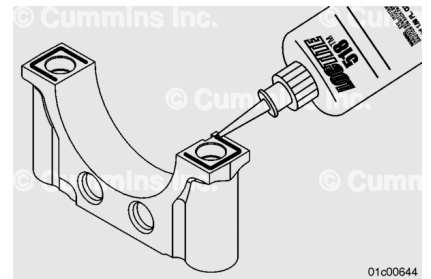
Push the crankshaft toward the rear of the engine to install the front thrust bearing and toward the front of the engine to install the rear thrust bearing.

⚠ CAUTION ⚠

Only use Loctite™ 518 thread sealant on the main cap joint mating surface. Other sealants can become hard, brittle and allow oil and debris into the main bearing/block joint, causing engine damage.

⚠ CAUTION ⚠

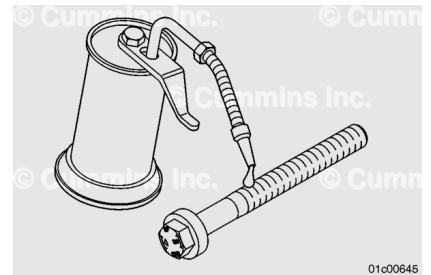
The bead must be 3 to 5 mm [0.12 to 0.20 in] wide and must not enter the main bearing shell inside the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.



Apply a u-shaped bead of Loctite™ 518 thread sealant to the two main bearing cap surfaces on each main bearing cap, as illustrated.

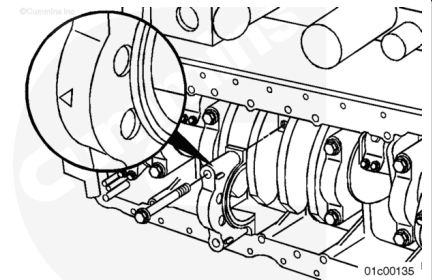
Use clean engine oil to lubricate the capscrew threads and head flange.

Drain the excess oil from the capscrews before installing them. Make sure there is **not** excessive oil in the threaded hole on the block.



⚠ CAUTION ⚠

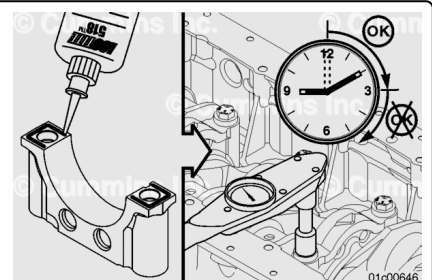
The main bearing caps are marked with a 'V'. Position the 'V' mark so that it points to the front of the engine. Incorrect positioning will result in severe engine damage.



Align the capscrew holes in the cap with the holes in the cylinder block. Make sure the lower bearing shell is in position.

Install the capscrews through the cap and into the cylinder block.

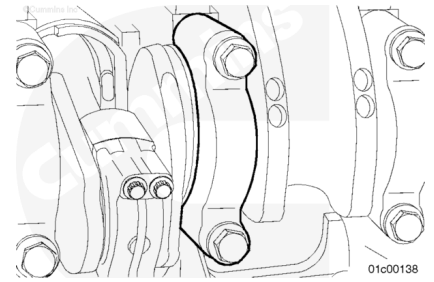
To reduce the possibility of bore size issues, clearance issues, or both, the main bearing capscrews **must** be tightened within 15 minutes of applying the Loctite™ 518 thread sealant.



Make sure the bearing shell does **not** move.

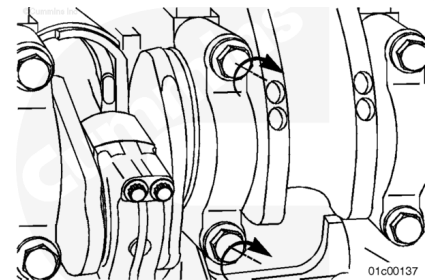
Only number 4 main bearing cap has thrust bearings. Make sure the thrust bearings are in place and properly seated when installing the number 4 main bearing cap.

Tighten the capscrews in far enough to use a heel bar to pry the caps into place.



Tighten both of the capscrews on a main bearing cap to 203 N•m [150 ft-lb].

Loosen both capscrews completely.

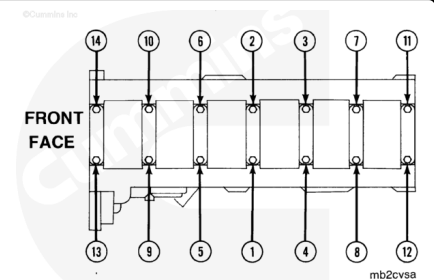


Use the torque plus angle method to tighten the main bearing capscrews in the sequence shown.

Torque Value:

1. 149 n•m [110 ft-lb]
2. Rotate 180 degrees.

It is recommended to use an air impact wrench and socket, Part Numbers 3163059 and 5573072, or equivalent, to rotate the capscrews the additional 180 degrees.

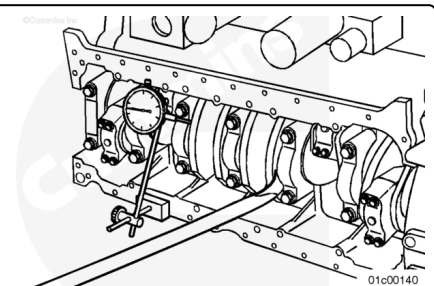


The crankshaft end clearance **must** be measured.

To measure the end clearance of the crankshaft, install a dial indicator onto the oil pan mounting flange.

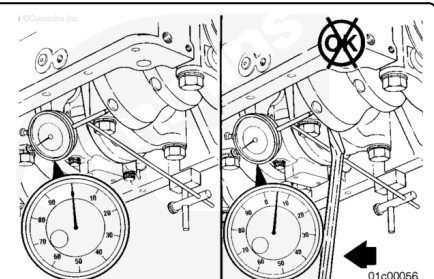
Place the tip of the gauge against the crankshaft counterweight.

Push the crankshaft toward the rear of the cylinder block.



Set the dial indicator at "0" (zero).

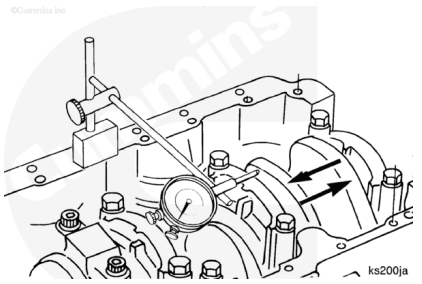
Push the crankshaft toward the front of the cylinder block.



Measure the crankshaft end clearance.

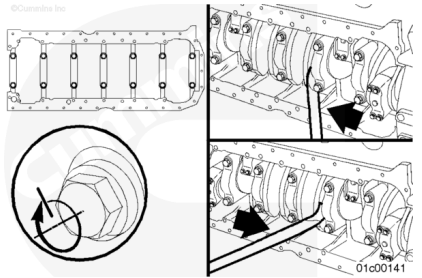
Crankshaft End Clearance - New or Reground

mm		in
0.10	MIN	0.004
0.50	MAX	0.020



If the end clearance is less than 0.10 mm [0.004 in], do the following:

1. Loosen all main bearing capscrews one turn.
2. Push the crankshaft toward the front and then toward the rear of the cylinder block.

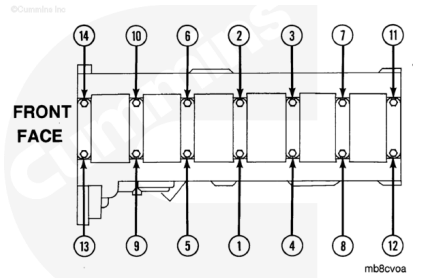


Tighten the main bearing capscrews in the sequence shown.

Torque Value:

1. 149 n•m [110 ft-lb]
2. Rotate 180 degrees.

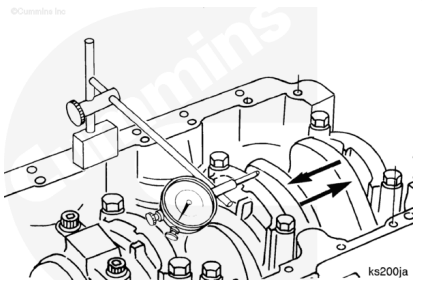
It is recommended to use an air impact wrench and socket, Part Numbers 3163059 and 5573072, or equivalent, to rotate the capscrews an additional 180 degrees.



Measure the crankshaft end clearance.

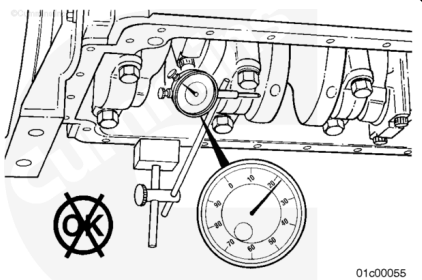
Crankshaft End Clearance - New or Reground

mm		in
0.10	MIN	0.004
0.50	MAX	0.020



If the crankshaft end clearance is still less than 0.10 mm [0.004 in], or more than 0.50 mm [0.020 in], the engine **must** be removed to allow for replacement of the crankshaft. Refer to Procedure 000-001 in Section 0.
(/qs3/pubsys2/xml/en/procedures/377/377-000-001.html)

If the crankshaft end clearance is still less than 0.10 mm [0.004 in], or more than 0.50 mm [0.020 in], the crankshaft **must** be replaced.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after any internal engine repairs to avoid internal component damage.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-089.html)
- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-031.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-035.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Prime the lubricating system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine until it reaches normal operating temperature and check for leaks and proper operation.